

Altitude Illness

(Mountain Sickness)

By **Andrew M. Luks**, MD, University of Washington

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Altitude illness occurs because of a lack of oxygen at high altitudes and affects mountain climbers, recreational hikers, skiers, and others who travel to high altitude.

- Symptoms include headache, tiredness, nausea or loss of appetite, irritability, and in more serious cases, shortness of breath, confusion, and even coma.
- Doctors diagnose altitude illness primarily based on the symptoms.
- Treatment may include rest, descending to a lower altitude, and sometimes medications, extra oxygen, or both.
- People may prevent these disorders by ascending slowly and sometimes by taking medications.

As altitude increases, the percentage of oxygen in air remains constant, but the atmospheric pressure decreases, thinning the air so that less oxygen is available. For example, compared with the air at sea level, the air at 19,000 feet (5,800 meters) contains only half the amount of oxygen. In Denver, which is located about 5,300 feet (1,615 meters) above sea level, the air contains 20% less oxygen.

Most people can ascend to 5,000 to 6,500 feet (1,500 to 2,000 meters) in 1 day without problems, but about 25% of people who ascend to 8,000 feet (2,500 meters) and 40% who ascend to 14,000 feet (4,340 meters) develop some form of altitude illness. Extremely high rates of these illnesses occur in religious pilgrims in the Himalayan mountains of Nepal and in climbers ascending Mount Kilimanjaro in Kenya and Tanzania. Both settings are known to attract climbers who ascend too rapidly, a primary risk factor for the development of altitude illness. Other factors that

contribute to developing altitude illness are the maximum altitude reached and the sleeping altitude.

The organs most commonly affected by altitude illness are the

- Brain (causing [acute mountain sickness](#) [AMS] and rarely [high-altitude cerebral edema](#) [HACE])
- Lungs (causing [high-altitude pulmonary edema](#) [HAPE])

Risk factors

The risk of developing altitude illness varies greatly among individuals. But generally, risk is increased in people who ascend to more than 2000 m (5000 to 6500 ft) and meet at least 1 of the following conditions:

- Have had previous altitude illness
- Live at sea level or very low altitude
- Go too high too fast
- Overexert themselves
- Sleep at too high an altitude

People who have disorders such as [asthma](#), [diabetes](#), [coronary artery disease](#), and mild [COPD \(chronic obstructive pulmonary disease\)](#) are not at increased risk for altitude illness. However, such people may have difficulties with these chronic medical problems at high altitude because of low blood oxygen levels (hypoxemia).

Asthma does not generally seem to be worse at high altitudes.

Spending less than a few weeks at higher altitudes (but below 10,000 feet [3,000 meters]) does not appear to be dangerous for a pregnant woman or the fetus.

Physical fitness does not protect against altitude illness.

Acclimatization

The body eventually adjusts (acclimatizes) to higher altitudes by increasing respiration, by producing more red blood cells to carry oxygen to the tissues, and by making other adjustments. Most people can adjust to altitudes of up to 10,000 feet (3,000 meters) in a few days. Adjusting to much higher altitudes takes many days or weeks, but some people can eventually carry out nearly normal activities at altitudes above 17,500 feet (about 5,300 meters). However, no one can fully acclimatize to long-term residence above that altitude.

Did You Know...

- Symptoms of acute mountain sickness (AMS) may be mistaken for a hangover, physical exhaustion, a migraine, or a viral illness.

Symptoms of Altitude Illness

Acute mountain sickness (AMS)

AMS is a mild form of altitude illness and is its most common form. It usually does not develop unless altitude is at least 8,000 feet (about 2,400 meters), but it can develop at lower altitudes in highly susceptible people. Symptoms usually develop within 6 to 10 hours of ascent (increase in altitude) and often include headache and 1 or more other symptoms, such as light-headedness, loss of appetite, nausea, vomiting, fatigue, weakness, or irritability. Some people describe the symptoms as similar to those of a hangover. Symptoms usually last 24 to 48 hours. Rarely, AMS progresses to a more severe form of altitude illness known as high-altitude cerebral edema.

High-altitude cerebral edema (HACE)

HACE is a rare but potentially fatal condition in which the brain swells with fluid. People with HACE have headache, confusion, and walking that is unsteady and uncoordinated (ataxia). If the disorder is not recognized and treated at an early stage, affected people may lapse into a coma. These symptoms may progress rapidly from mild to life-threatening within a few hours.

High-altitude pulmonary edema (HAPE)

HAPE is a fluid build-up in the lungs that usually develops 24 to 96 hours after a rapid ascent to over 8,000 feet (2,400 meters). It can occur in people even if they do not have symptoms of AMS. HAPE is responsible for most deaths due to altitude illness. People who live at high altitudes may develop a form of HAPE known as high-altitude resident HAPE, even if they do not descend to and then return from a lower altitude. People who live at high altitude and descend to lower elevation for, for example, a vacation, can develop pulmonary edema upon re-ascent to their residence, a phenomenon known as reentry HAPE.

Respiratory infections, even minor ones, may increase the risk of HAPE. Symptoms are worse at night when people lie down and can quickly become more severe if HAPE is not recognized and treated promptly. Mild symptoms usually include a dry cough and

shortness of breath after only mild exertion. Moderate symptoms include shortness of breath at rest and a bluish tinge to the skin, lips, and nails ([cyanosis](#)). Severe symptoms include gasping for breath, pink or bloody sputum, severe cyanosis, and making gurgling sounds while breathing. HAPE may worsen quickly and result in respiratory failure, coma, and death within a few hours.

Other symptoms

Swelling of the hands, the feet, and, on awakening, the face is common. The swelling causes little discomfort and usually goes away in a few days or with descent.

Headache, without any other symptoms of acute mountain sickness, is also common.

Retinal hemorrhages (small areas of bleeding in the retina at the back of the eye) can develop after ascent to altitudes above 9,000 feet (2,700 meters). These hemorrhages are common above 16,000 feet (4,800 meters). People usually have no symptoms unless the hemorrhage occurs in the part of the eye that is responsible for central vision (the macula). In such cases, people may notice a small blind spot without eye pain. Retinal hemorrhages resolve over a period of weeks without causing long-term problems. People who develop blind spots in their vision while climbing or trekking at high altitude should descend to lower elevation and seek further evaluation. Re-ascent to high altitude can be undertaken once the hemorrhage has resolved.

Diagnosis of Altitude Illness

- A doctor's evaluation
- For high-altitude pulmonary edema (HAPE), a chest x-ray and blood oxygen levels, if available

Doctors diagnose altitude illness based mainly on the symptoms. In people with HAPE, doctors can usually hear fluid in the lungs through a stethoscope. An x-ray of the chest and measurement of the amount of oxygen in the blood can help confirm this diagnosis.

A head CT or MRI of the brain can be used to help confirm the diagnosis of HACE but is generally not necessary.

What Is Chronic Mountain Sickness?



Most altitude sickness occurs in people who quickly ascend to high altitude. But some people develop altitude-related diseases only after living a long time at high altitude.

Chronic mountain sickness (Monge's disease) is a disease that develops in some people who live at altitudes higher than 10,000 feet (about 3,000 meters) for many months or years. Symptoms include fatigue, shortness of breath, aches and pains, and a blue color to the lips and skin (cyanosis). In affected people, the body overcompensates for the lack of oxygen by overproducing red blood cells. The extra red blood cells make the blood so thick that it may become difficult for the heart to pump enough blood to the organs throughout the body.

Periodic removal of blood (phlebotomy) provides temporary relief, but the most effective treatment is descent to a low altitude. Sometimes acetazolamide also provides relief. Complete recovery can take months. People need to remain at low altitude.

Monge's disease is common in the Andes Mountains but has also been seen in high-altitude communities in Colorado. In other areas of the world (for example, in Tibet), some lowlanders who have relocated to high elevation develop a different form of chronic altitude illness marked by high blood pressures in the lungs and decreased function of the right side of the heart, without overproduction of red blood cells.

Treatment of Altitude Illness

Descent to lower elevation is the best treatment for all forms of acute altitude illness.

- For mild symptoms of acute mountain sickness (AMS), stopping the ascent (for example, stopping the climb to higher altitude) and treating symptoms with medications
- For severe or slow-to-resolve acute AMS, descent to a lower altitude and treatment with medications
- For high-altitude cerebral edema (HACE) and high-altitude pulmonary edema (HAPE), immediate descent to a low altitude and medications (if

descent is not feasible, medications and supplemental oxygen or a portable hyperbaric chamber)

People who have swelling of the hands, feet, and face but no other symptoms of altitude illness do not need treatment. The swelling goes away on its own after a few days or following descent. Poor sleep is a common problem at high altitude, even among healthy people, and, by itself, is not a reason to descend to lower elevation.

People with mild AMS must stop their ascent and rest. They should not ascend to higher altitudes until symptoms disappear. Other treatment includes fluids and acetaminophen or nonsteroidal anti-inflammatory drugs (NSAIDs), such as ibuprofen, to help relieve headache. Fluids do not treat AMS, but do eliminate dehydration, the symptoms of which can imitate those of AMS. Most people with AMS improve within 1 or 2 days. Sometimes, acetazolamide or dexamethasone is given to help relieve symptoms.

If AMS symptoms are more severe, or if symptoms persist or worsen despite treatment, the person should descend to a lower altitude, preferably 1,650 to 3,200 feet (500 to 1,000 meters) lower. The descent often provides rapid relief. The person is also given fluids, acetaminophen or NSAIDs, and acetazolamide or dexamethasone.

If HACE develops, the person should immediately descend as far down as possible. Oxygen and dexamethasone should be taken. Acetazolamide may be added.

People with HAPE should immediately descend to a low altitude. Oxygen should be given if it is available. Nifedipine may temporarily help by decreasing blood pressure in the arteries to the lungs. Heavy exertion should be avoided during descent as this can worsen pulmonary edema. People who promptly descend usually recover from HAPE within 24 to 48 hours. Most people who develop HAPE are admitted to a hospital, but some (for example, those in areas with adequate medical resources such as a ski resort community) can be discharged with supplemental oxygen if they have family or friends who can monitor them.

When prompt descent to a lower altitude is not possible and people are seriously ill with severe AMS, HACE, or HAPE, a hyperbaric bag can be used to buy time. This device consists of a lightweight, portable fabric bag large enough to completely contain a person and a manually operated pump. The person is sealed tightly in the bag, and the bag's internal pressure is then increased using the pump. The increased air pressure simulates a decrease in altitude. The person remains in the bag until symptoms resolve. The hyperbaric bag is as beneficial as supplemental oxygen, which often is not available when mountain climbing, but is not a substitute for descent.

Prevention of Altitude Illness

- Slow ascent (increase altitude slowly)
- Sometimes [acetazolamide](#) or [dexamethasone](#)

Rate of ascent

The best way to prevent altitude illness is to ascend slowly. The altitude at which a person sleeps is more important than the maximum height reached during the day. Control of the rate of ascent (called graded ascent) is essential for activity any higher than 8,000 feet (2,400 meters). Above 10,000 feet (3,000 meters), climbers or trekkers should not increase their sleeping altitude by more than 1,600 feet (500 meters) per day and should include a rest day (sleep at the same altitude) every 3 to 4 nights before they sleep at any higher altitudes. If people cannot limit each day's ascent to less than 1,600 feet (500 meters), they should limit their average ascent over the entire ascent to less than 1,600 feet (500 meters) per day. This may require adding rest days. During rest days, day hikes to higher elevations are acceptable as long as people return to the lower level for sleep.

People vary in their ability to ascend without developing symptoms. Thus, a climbing party should be paced for the member who acclimatizes to high altitude the slowest.

Acclimatization reverses quickly. If acclimatized people have descended to low levels for more than a few days, they must once more follow a graded ascent when they reascend.

Medications

[Acetazolamide](#), which can be started the night before the ascent, can reduce the likelihood of altitude illness. If taken after illness has begun, [acetazolamide](#) may help lessen symptoms. [Acetazolamide](#) should be stopped when descent is initiated or after a few days have been spent at the peak elevation. [Dexamethasone](#), an alternative to [acetazolamide](#), can also reduce the likelihood of acute mountain sickness and treat its symptoms.

Taking analgesics such as [nonsteroidal anti-inflammatory drugs](#) (NSAIDs) may help prevent high-altitude headache.

People who have had previous episodes of high-altitude pulmonary edema should be alert for any symptoms of a recurrence and descend immediately if symptoms occur.

Some doctors also recommend such people take nifedipine or tadalafil by mouth as a preventive measure.

General measures

Avoiding strenuous exertion for 1 or 2 days after arrival may help prevent altitude illness. Heavy alcohol consumption, opioids, and sedatives should be avoided, particularly shortly before sleep. Habitual caffeine drinkers should be aware of the possibility of caffeine-withdrawal headaches if they stop consuming caffeine on their excursion.

Although physical fitness enables greater exertion at altitude, it does not protect against any form of acute altitude illness. Acetazolamide can be used to improve sleep, which is disturbed for many people traveling to high altitude.

Drug Information for the Topic



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